

Design for class “PlaceOrderController”

Inheritance of class “BaseController”

PlaceOrderController
- currentOrder : Order - payOrderController : PayOrderController - emailSender : EmailSender
+ requestToPlaceOrder(cart : Cart) : void + submitDeliveryInfo(info : HashMap<String,String>) : void + requestPayment(method : String) : void - checkStock(cart : Cart) : boolean - validateDeliveryInfo(info : HashMap<String,String>) : DeliveryInfo - calculateShippingFee(info : DeliveryInfo) : void - createOrder(cart : Cart, info : DeliveryInfo) : void - saveOrder() : void - sendConfirmationEmail() : void - handleStockError(message : String) : void - handleInvalidDeliveryInfo(message : String) : void

Table 1. Attribute Design

#	Name	Data type	Default value	Description
1	currentOrder	Order	null	The order currently being created or processed.
2	payOrderController	PayOrderController	null	Controller responsible for handling the payment sub-flow.

Table 2. Operation Design

#	Name	Return type	Description / Purpose
1	requestToPlaceOrder(cart)	void	Start the order placement process by validating cart and checking stock.
2	submitDeliveryInfo(info)	void	Validate delivery info and attach it to the order.
3	requestPayment(method)	void	Initiate payment using the specified method (QR/PayPal).

4	checkStock(cart)	boolean	Check if each product in cart has sufficient stock.
5	validateDeliveryInfo(info)	DeliveryInfo	Validate and convert raw input info into a DeliveryInfo object.
6	calculateShippingFee(info)	void	Compute delivery fee based on weight, province, or policy.
7	createOrder(cart, info)	void	Create an Order object and assign subtotal, VAT, items.
8	saveOrder()	void	Persist the newly created order to the database.
9	sendConfirmationEmail()	void	Send an email notifying the user of the successful order.
10	handleStockError (message)	void	Present user with stock-related errors.
11	handleInvalidDeliveryInfo (message)	void	Present user with delivery-information errors.

Method Design:

1. requestToPlaceOrder

- Parameter:
 - cart: Cart - the user's current cart.
- Exception:
 - OutOfStockException if some products lack sufficient stock.
- Method:
 - Call checkStock(cart).
 - If false → call handleStockError(message).
 - If true → proceed to construct delivery info screen.
 - Store cart temporarily or prepare draft order.

2. submitDeliveryInfo

- Parameter:

- info: HashMap<String, String> — raw input fields from DeliveryInfoScreen.
- Exception:
 - InvalidDeliveryInfoException if any field is invalid or missing.
- Method:
 - Call validateDeliveryInfo(info) to get a DeliveryInfo object.
 - Call calculateShippingFee(deliveryInfo) to compute the shipping fee.
 - Call createOrder(cart, deliveryInfo) to form the full order.
 - Navigate to InvoiceScreen.

3. requestPayment

- Parameter:
 - method: String — e.g., "QR" or "PAYPAL".
- Exception:
 - UnsupportedPaymentMethodException if method is not recognized.
- Method:
 - Call payOrderController.processPayment(currentOrder, method).
 - If success → save order & send confirmation email.
 - Navigate to SuccessfulOrderScreen.
 - If failure → show appropriate error message.

4. checkStock

- Parameter:
 - cart: Cart
- Exception:
- Method:
 - For each OrderItem in cart:
 - Call product.checkStock(quantity).
 - Return true only when all items satisfy stock requirement.

5. validateDeliveryInfo

- Parameter:

- info: HashMap<String, String>
 - Return: DeliveryInfo
- Exception:
 - InvalidDeliveryInfoException if missing fields or wrong format.
- Method:
 - Validate name, phone number, address, province.
 - Construct and return DeliveryInfo object.

6. calculateShippingFee

- Parameter:
 - info: DeliveryInfo
- Exception:
- Method:
 - Compute shipping fee based on weight, province, or rule.
 - Assign to info.fee.

7. createOrder

- Parameter:
 - cart: Cart
 - info: DeliveryInfo
- Exception:
- Method:
 - Instantiate currentOrder.
 - Copy items from cart to orderItems.
 - Compute subtotal, vat, total.
 - Attach DeliveryInfo to order.

8. saveOrder

- Parameter:

- Exception:
 - DatabaseWriteException if saving fails.

- Method:
 - Persist order to database.
 - Update product stock.

9. sendConfirmationEmail

- Parameter:

- Exception:
 - EmailSendFailureException

- Method:
 - Build email content summarizing the order.
 - Invoke email subsystem to send.

10.handleStockError

- Parameter:
 - message: String

- Exception:

- Method:
 - Display error dialog.
 - Ensure user remains on CartScreen.

11.handleInvalidDeliveryInfo

- Parameter:
 - message: String

- Exception:

- Method:
 - Show invalid info message on DeliveryInfoScreen.

